# test3 torques, momentum, center of mass

This is a preview of the published version of the quiz

Started: Apr 29 at 7:48pm

## Quiz Instructions

**One attempt. You can not see your answers. select wisely. don't rush. no make up. Use tablet or laptop**

**No make-up. No complain**

**Remember: momentum is a vector. In 1D, it can be positive or negative. It is negative if the velocity is negative (left or down)**

Question 1 5 pts

The lug nuts on a car wheel require tightening to a **torque** of 90 N · m. If a 30-cm long wrench is used, what is the magnitude of the *minimum* force required using the wrench?

hint: convert

- 150 N
- 300 N
- 15 N
- 30 N

## Question 2 5 pts

The famous Leaning Tower of Pisa doesn't topple over because its center of gravity is

- above a place of support.
- in the same place as its center of mass.
- stabilized by its structure.
- relatively low for such a tall building.
- displaced from its center.

## Question 3 5 pts

A 0.10-kg ball, traveling horizontally at 25 m/s, strikes a wall and rebounds at 19 m/s. What is the magnitude of the change in the momentum of the ball during the rebound?

- 5.4 kg · m/s
- 1.8 kg · m/s

○ 1.2 kg · m/s

○ 72 kg · m/s

○ 4.4 kg · m/s

## Question 4 5 pts

A 1000-kg whale swims horizontally to the right at a speed of 6.0 m/s. It suddenly collides directly with a stationary seal of mass 200 kg. The seal grabs onto the whale and holds tight. What is the momentum of these two sea creatures just after their collision? You can neglect any drag effects of the water during the collision.

(hint: its like a collision between 2 cars that stick to each other- hit and stick). I am asking for the total momentum after the collision

○ 2000 kg · m/s

○ 1200 kg · m/s

○ 7200 kg · m/s

○ 6000 kg · m/s

○ 0.00 kg · m/s

## Question 5 5 pts

A uniform 40-N board supports two children weighing 500 N and 350 N. If the support is at the center of the board and the 500-N child is 1.5 m from its center, how far is the 350-N child from the center?

○ 1.5 m

○ 1.1 m

○ 2.1 m

○ 2.7 m

## Question 6 5 pts

A 328-kg car moving at 19.1 m/s in the +x direction hits from behind a second car moving at 13.0 m/s in the same direction. If the second car has a mass of 790 kg and a speed of 15.1 m/s right after the collision, what is the velocity of the first car after this sudden collision?

use conservation of momentum. they don't stick

- 18.2 m/s
- 14.0 m/s
- 24.2 m/s
- -14.0 m/s

Question 7 5 pts

Granny and junior are facing each other on an ice ring. They are wearing ice skates and we neglect friction. Granny is 200 pounds and junior 50 pounds. Granny pushes Junior who moves back at a speed of 2 m/s. What is the speed of Granny (action-reaction and conservation momentum)

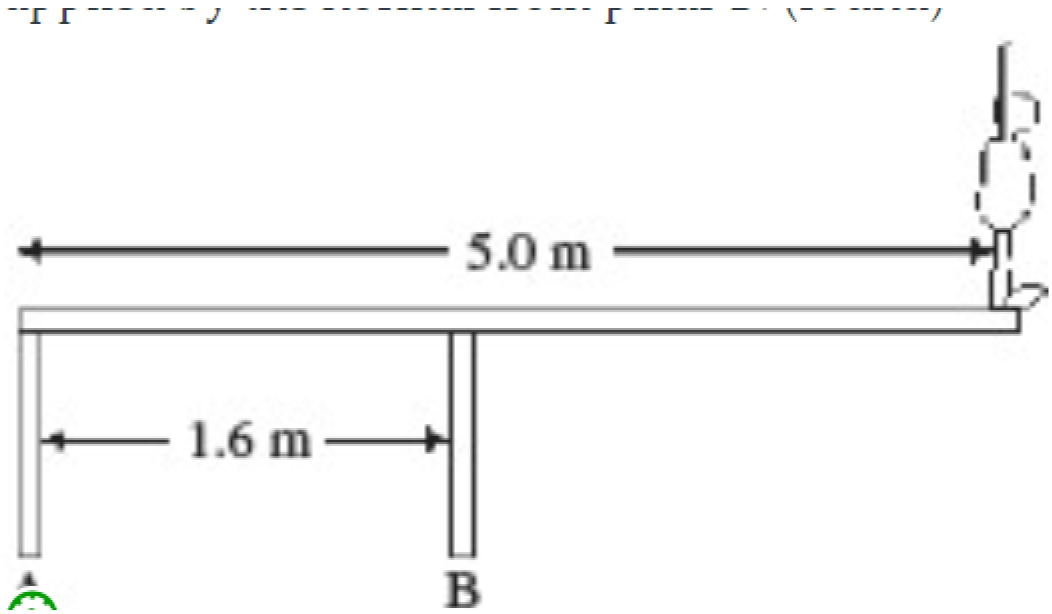
obviously a recoil situation

- 0.5m/s
- 2m/s
- 4m/s
- 0
- 8m/s
- 2m/s

Question 8 5 pts

An 82-kg diver stands at the edge of a light 5.0-m diving board, which is supported by two vertical pillars that are 1.6 m apart, as shown in the figure. Find the magnitude and direction of the force exerted by pillar B (normal force).

hint: take the fulcrum at A . 3 forces are acting on the board. the weight of the diver (down) . the normal force from pillar B (up) and the normal force from pillar A(up). If the fulcrum is at A , you only consider the torque applied by the weight of the diver and the torque applied by the normal from pillar B. (round)




- 2500N up
- 1500N up
- 800N up
- 500N up

### Question 9 5 pts

A man in a gym is holding an 8.0-kg weight at arm's length, a distance of 0.55 m from his shoulder joint. What is the torque about his shoulder joint due to the weight if his arm is horizontal?

- 15 N · m
- 43 N · m
- 0 N · m
- 0.24 N · m
- 4.4 N · m

Question 10 5 pts

A 1200-kg cannon suddenly fires a 100-kg cannonball at 35 m/s. What is the recoil speed of the cannon? Assume that frictional forces are negligible and the cannon is fired horizontally.

- 3.2 m/s
- 3.5 m/s
- 2.9 m/s
- 35 m/s

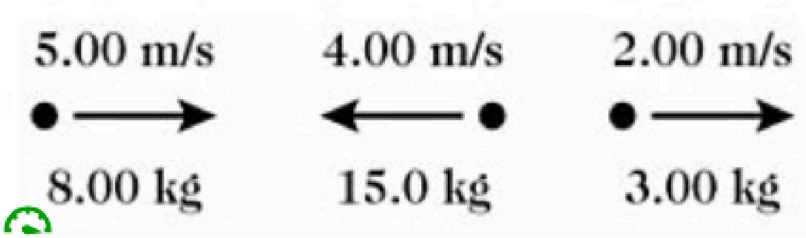
Question 11 5 pts

If you place a pipe over the end of a wrench when trying to rotate a stubborn bolt, effectively making the wrench handle twice as long, you'll multiply the torque by

- two.
- eight.
- four.

Question 12 5 pts

Three objects are moving along a straight line as shown in the figure. Taking the positive direction to be to the right, what is the total momentum of this system?




hint: momentum is a vector. It can be + or - in 1D

hint: momentum is a vector. It can be + or - in 1D

0.00 kg · m/s

-106 kg · m/s

-14.0 kg · m/s

+106 kg · m/s

+14.0 kg · m/s

Question 13 5 pts

You pull a wagon with a force of 40N over a distance of 1m.

The force makes an angle of 60 degrees with the positive x-axis. What is the work that you are doing.

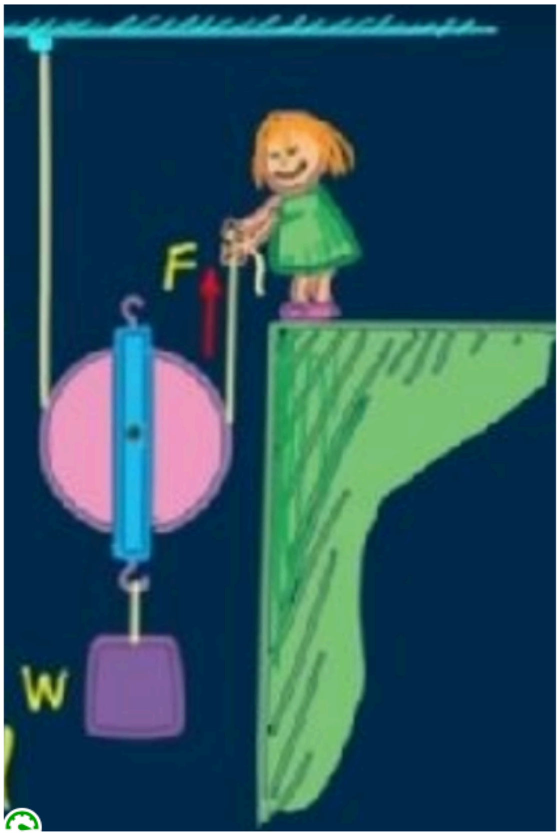
20N.m

50 N.m

60N.m

200N.m

Question 14 5 pts

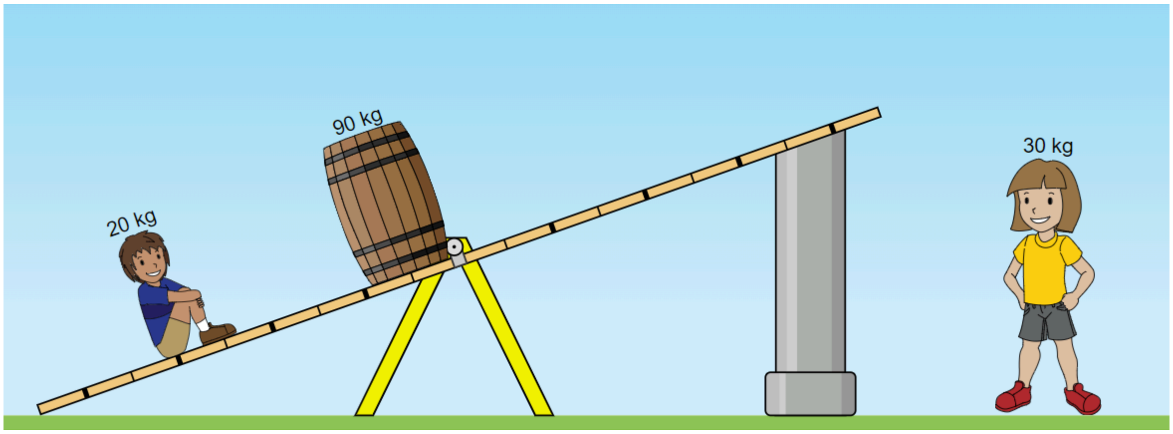



The load (W) is 400N. And the load (in purple) moves up by 1m up (at a constant speed). What is true?

Suzie is pulling at  a constant speed the load

- ◯ Suzie pulls with a force of 200N over a distance of 1m
- ◯ Suzie pulls with a force of 400N over a distance of 2m
- ◯ Suzie pulls with a force of 200N over a distance of 2m
- ◯ Suzie pulls with a force of 400N over a distance of 1m

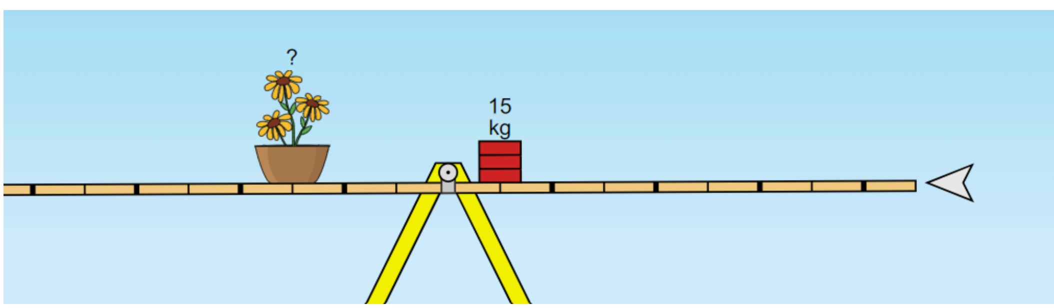
Question 15 10 pts




To get balance, little girl needs to sit at the _____ th mark away from the fulcrum

- 6th
- can not be balanced
- 8th mark
- 7th
- 5th

Question 16 5 pts




What is the mass of the flower pot

- 10kg
- 5kg
- 15kg
- 20kg

Question 17 5 pts

The 95kg red guy is moving at a speed of 6m/s at right and the blue 110kg is moving at a speed of 4m/s @ left.

They collide and stick to each other. What happens next


○ they collide and stop abruptly

○ can not be determined. not enough info

○ They move to the right

○ they move to the left

Question 18 5 pts

Building on the previous question. The magnitude of the velocity of the 2 players after the collision is about

○ 9.5m/s

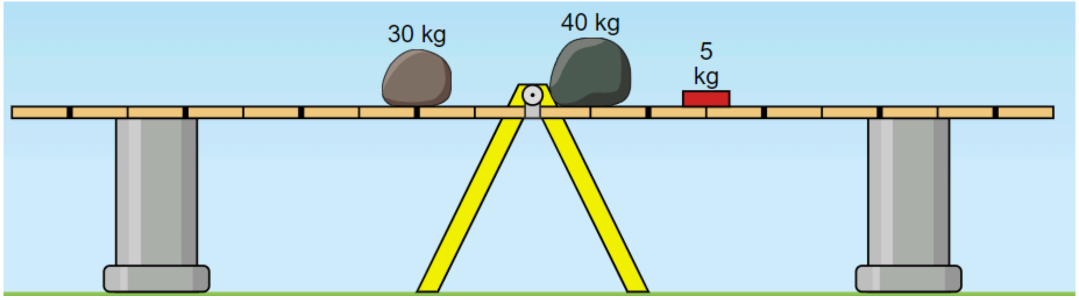
○ 6m/s

○ 0.6m/s

○ 4m/s

Question 19 5 pts

what will happen when you remove the poles




- [ ] It will topple to the left (not balanced)
- [ ] it will topple to the right (not balanced)
- [ ] it will be balanced
- [ ] Ca not be determined

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